Floating Points

Class-01

Floating Point Arithematic

- $330.330 = 33.0330 \times 10^{1} = 3.30330 \times 10^{2}$

  Fixed Point — Floating Number

- IEEE double Precision.

Normalized Convention / Scientific Notation

$58.73 \times 10^{2} = 5.873 \times 10^{3}$

$0.000325 \times 10^{2} = 3.25 \times 10^{-4+2}$

$= 3.25 \times 10^{-2}$

Scientific Notation

Binary Base, $\beta = 2 (0, 1)$; digit $\{0, 1\}$

e = exponential.
m = mantissa; how many digits are after (.)

The Formula,

$$\pm (1 . d_1 \, d_2 \, d_3 \ldots d_m)_{\beta} \times \beta^{e}$$

Main Rule One Non-zero (1) digit on the left side of the decimal point.

Example

$\#(110.1110)_2 = (110.111)_2$

$= (110111)_2 \times 2^{-3}$

$= (55)_{10} \times 2^{-3}$

Normalized, Base, $\beta = 2$

mantissa, $m = 2$

exponent, $e = \{2, 3\}$

Answer

$$\pm (1.\square\square)_2 \times 2^{\square}$$

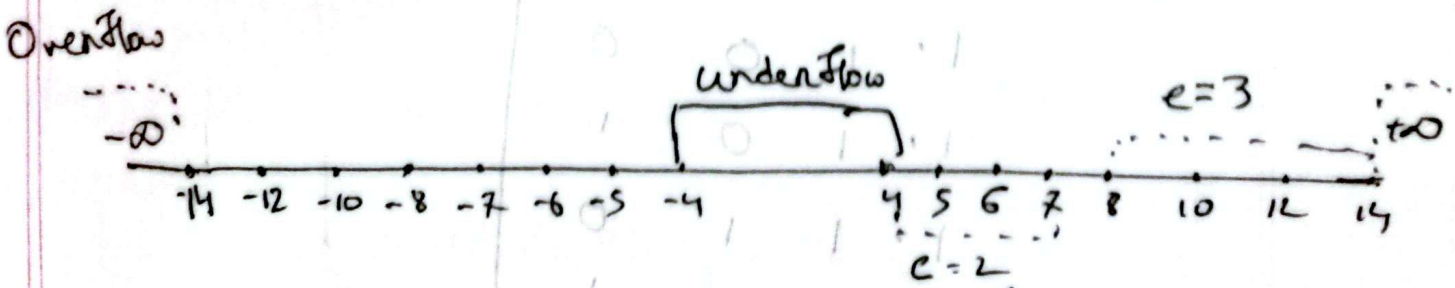
(each $\square$: 2 bits; exponent $\square$: 2 bits)

8×2 = 16 combinations possible

[2×2×2×2 = 16]

			2^v	2^3
$(1)_{10}$ ←	1.	0	0	2^v	2^3
$(1.25)_0$ ←	1.	0	1	2^v	2^3
$(1.5)_{10}$ ←	1.	1	0	2^v	2^3
$(1.75)_0$ ←	1.	1	1	2^v	2^3

$(1)_{10} + 2^v$	$2^3 \times (1)_{10}$
4	8
5	10
6	12
7	14



-0 represents -14; +0 represents +14; 0·(-4 to 4)

-14> and 14²<; any val other than -14 < x < 14 Overflow.

largest number ⟶ $+1.11 \times 2^3$

smallest number ⟶ -1.00×2^2

smallest number ⟶ $+1.00 \times 2^2$ [Non-Neg]

Normalized Base $\beta = 2$

mantissa $m = 3$

exponent, $e = \{x : -2 \le x \le 3 \text{ and } x \in \mathbb{N}\}$

or, $e \Rightarrow e_{max} = 3$; $e_{min} = -2$

Here, $\beta = 2$, $m = 3$, $e = \{-2, -1, 0, 1, 2, 3\}$

So, $\pm (1.\square\square\square)_2 \times 2^{\square}_{6}$

2^{-2}	2^{-1}	2^0	2^1	2^2	2^3
1.0	0	0			
1.0	0	1			
1.0	1	0			
1.0	1	1			
1.1	0	0			
1.1	0	1			
1.1	1	0			
1.1	1	1			

So, with neg consideration	without neg consideration
$max = (1.1111)_2 \times 2^3$	$max = (1.111)_2 \times 2^3$
$min = -(1.111)_2 \times 2^3$	$min = (1.000)_2 \times 2^{-2}$
total: $2 \times 2 \times 2 \times 2 \times 6$	total: $2 \times 2 \times 2 \times 6$

Class-2

We have 3 Format / Convention

① Standard Format

$$\pm (0.d_1 d_2 d_3 \ldots d_m)_\beta \times \beta^e$$

Here, d_1 is always Non-zero. So, $d_1 = 1$

② Normalized Format

$$\pm (1.d_1 d_2 d_3 \ldots d_m)_\beta \times \beta^e$$

③ Denormalized Format

$$\pm (0.1 d_1 d_2 d_3 \ldots d_m)_\beta \times \beta^e$$

Here Base, $\beta = 2$, Mantissa, $m = 3$

exponent, $e_{min} = -2$; $e_{max} = 3$, $e = \{-2, -1, 0, 1, 2, 3\}$

Denormalized

$$\pm (0.1 \underset{2}{_} \underset{2}{_} \underset{2}{_})_2 \times 2^{\square}$$

	Only Positive	Both Pos & Neg
min	$+ (0.1000)_2 \times 2^{-2}$	$- (0.1111) \times 2^3$
max	$+ (0.1111)_2 \times 2^3$	$+ (0.1111)_2 \times 2^3$
total	$2 \times 2 \times 2 \times 6$	$2 \times 2 \times 2 \times 2 \times 6$

Standard

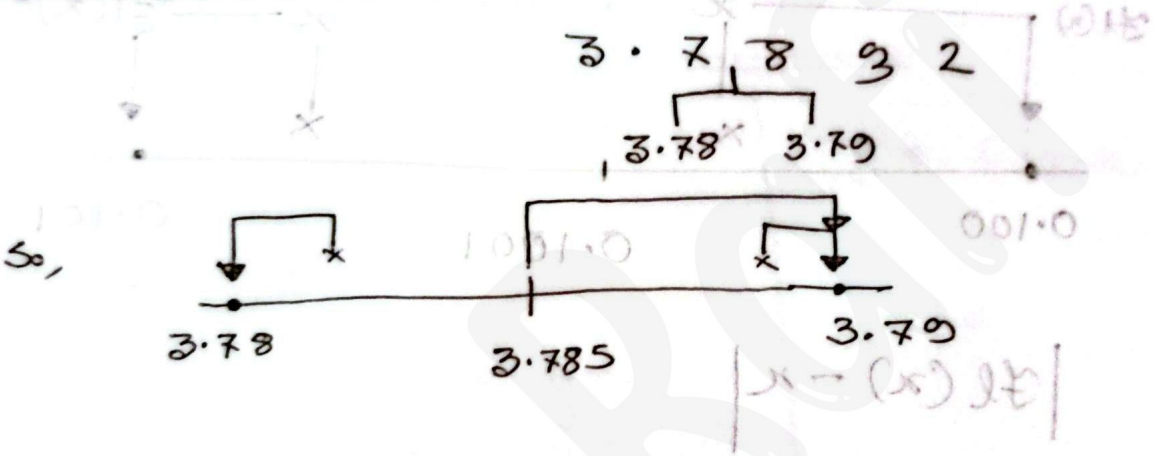
$$\pm (0.1 \underset{2}{_} \underset{2}{_})_2 \times 2^{\square}$$

	Only Positive	Both pos & Neg
min	$+ (0.100)_2 \times 2^{-2}$	$- (0.111)_2 \times 2^3$
max	$+ (0.111)_2 \times 2^3$	$+ (0.111)_2 \times 2^3$
total	$2 \times 2 \times 6$	$2 \times 2 \times 2 \times 6$

Rounding

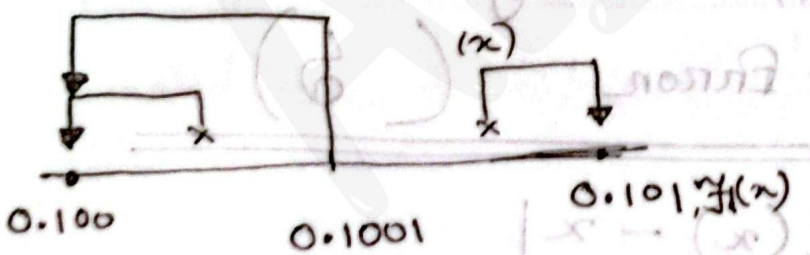
Rounding Error

■ Decimal Rounding



■ Binary Rounding

Rounding up to the even side.





Middle point হলে রাখার way ↓

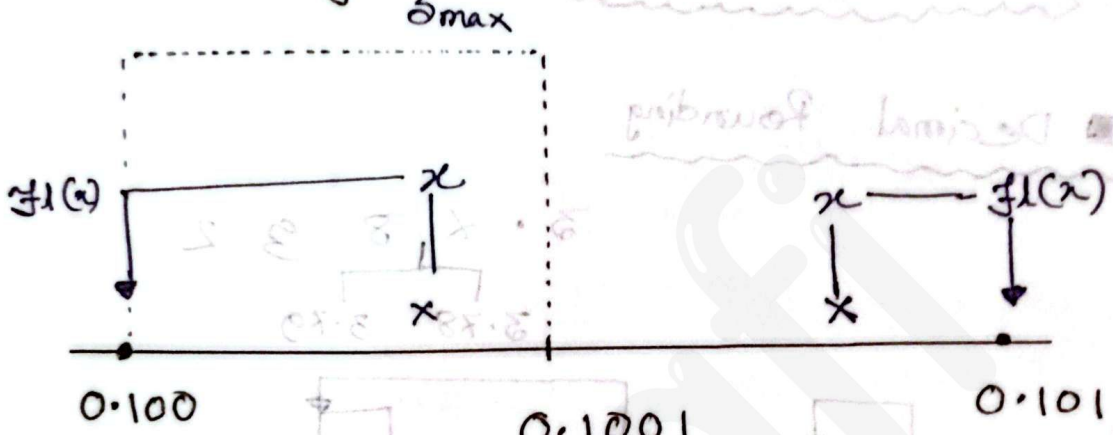
Previous number এর শেষে 1 add করে দিব।

x → fl(x) ↓

Rounded value of x = fl(x)

- Absolute Rounding Error $|fl(x) - x|$

  - It doesn't give us an idea of a scale.



$$|fl(x) - x|$$

- Relative Rounding Error, Inherent Scale Invalid/∧ Rounding Error, Percentage Error (δ)

$$\delta = \frac{|fl(x) - x|}{|x|}$$

Maximized δ,

Machine Epsilon

$\delta_{max} = \varepsilon_m$; x এর middle point এ থাকলে তখন error max হবে।

Definition of Rounding function

$$\delta = \frac{|fl(x) - x|}{|x|}$$

$\delta \uparrow$ Error $\uparrow$

$$\Rightarrow \delta x = fl(x) - x$$

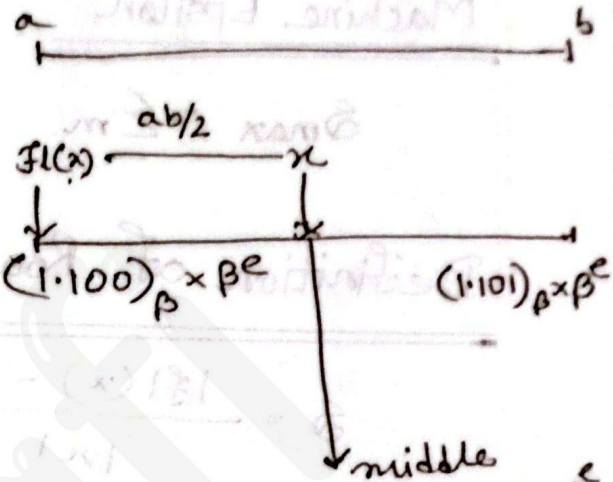
$$\Rightarrow fl(x) = \underset{\text{actual val}}{x} + \underset{[0,1]}{\delta x}$$

Proof of Machine Epsilon / δ_{max} / ε_m

maximization of $\delta / max(\delta) / \delta_{max} = \frac{|fl(x) - x|}{|x|}$ (max / min)

Normalized Approach

Let us take 2 consecutive points.



$(1.100)_\beta \times \beta^e$ ... (ii)

$(1.101)_\beta \times \beta^e$ ... (i)

(i) - (ii) $\Rightarrow$ $(0.001)_\beta \times \beta^e$

So, $ab = 1 \times \beta^{-m} \times \beta^e$

$= \beta^{-m} \times \beta^e$

Now,

$$\delta_{max} = \varepsilon_m = \frac{\frac{1}{2} ab}{\beta^e}$$

So, $\varepsilon_m = \frac{\frac{1}{2}\beta^{-m} \times \beta^e}{\beta^e}$

So, $\varepsilon_m = \frac{1}{2}\beta^{-m}$

So, $\epsilon = \frac{|fl(x) - x|}{|x|}$

=

So, minimum number in Normalized with $m = 3$,

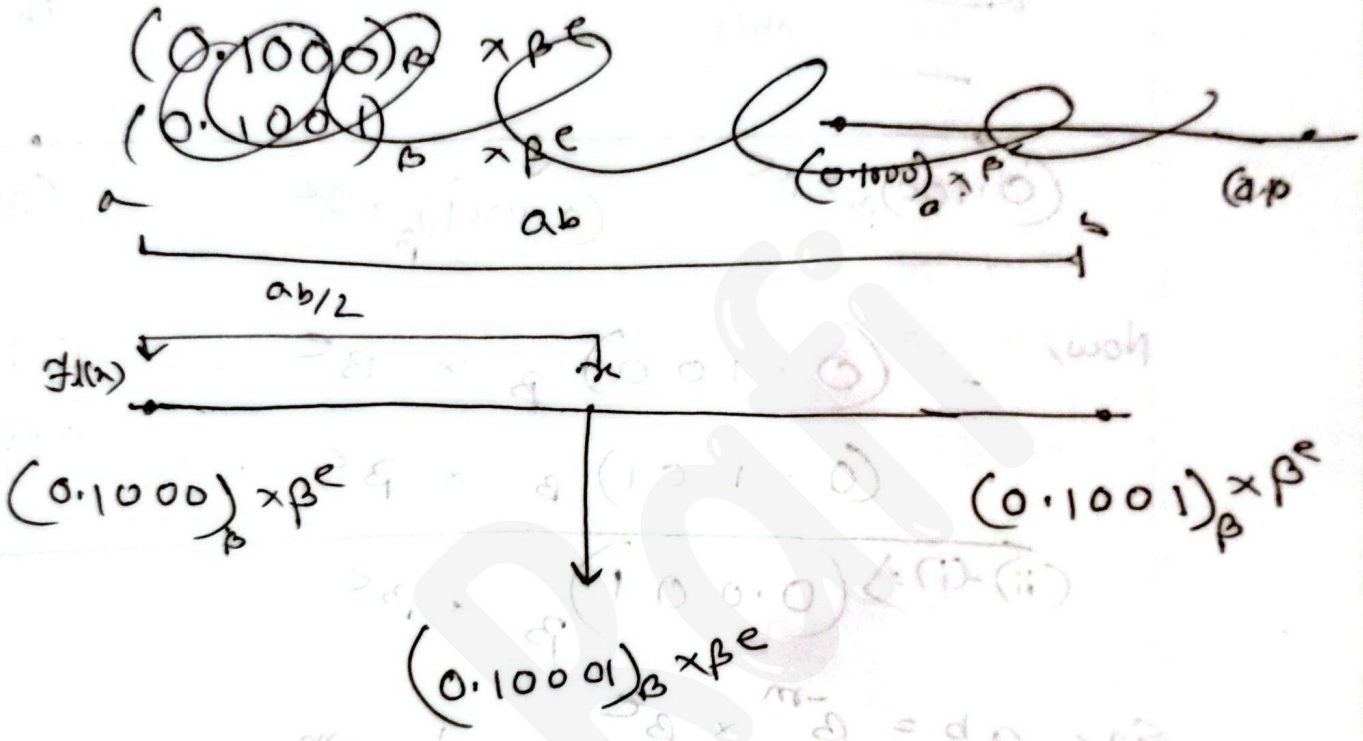
$(1.00)_\beta \times \beta^e$

$= \beta^0 \times \beta^e$ [as 1 Here, stays in the index 0]

$= 1 \times \beta^e = \beta^e$; $|x|_{min} = \beta^e$

Denormalized Approach

$m = 3$



So, $-(0.1000)_\beta \times \beta^e$ ... (i)

$(0.1001)_\beta \times \beta^e$ ... (ii)

(ii)-(i) $\Rightarrow$ $(0.0001)_\beta \times \beta^e$

So, $ab = \beta^{-m+1} \times \beta^e$

So, $\delta_{max} = \varepsilon_m = \dfrac{\frac{1}{2}\beta^{-m+1} \times \beta^e}{\beta^e \times \beta^{-m+1}}$

So, $\varepsilon_m = \frac{1}{2}\beta^{-m}$

$ab = \frac{1}{2}\beta^{-m+1}\beta^e$ [as we've an extra bit]

So, $\varepsilon_m = \dfrac{\frac{1}{2}\beta^{-m+1}\beta^e}{\beta^{-1}\beta^e} = \frac{1}{2}\beta^{-m}$

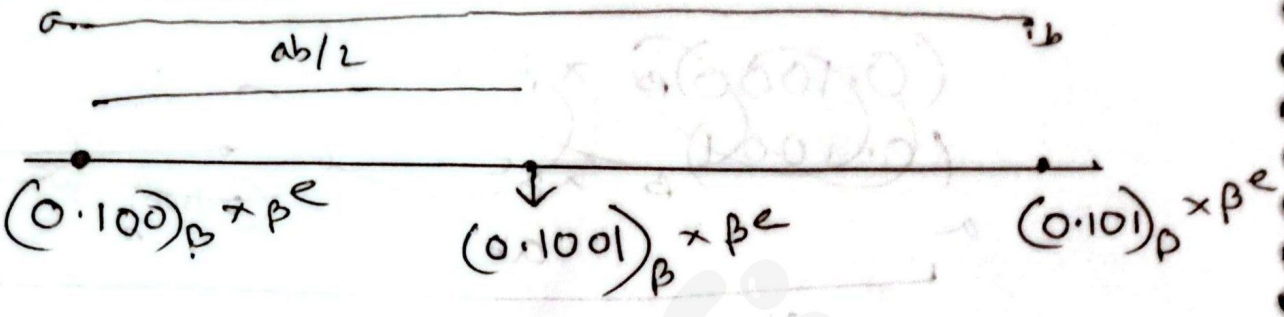
min number will be,

$(0.1000)_\beta \times \beta^e$

$b = \beta^{-1} \times \beta^e$; [1 in (-1) pos]

So, $|x|_{min} = \beta^{-1} \times \beta^e$

Standard approach) $m = 3$



Now, $(0.100)_\beta \times \beta^e$ ... (i)

$(0.101)_\beta \times \beta^e$ ... (ii)

(ii) - (i) $\Rightarrow (0.001)_\beta \times \beta^e$

So, $ab = \beta^{-m} \times \beta^e$

So, $\delta_{max} = \varepsilon_m = \frac{1}{2}\beta^{-m} \times \beta^e$

$= \frac{1}{2}\beta^m \beta^e$

Now, min number will be

$(0.100)_\beta \times \beta^e$

$= \beta^{-1}\beta^e$; So, $|x|_{min} = \beta^{-1}\beta^e$.

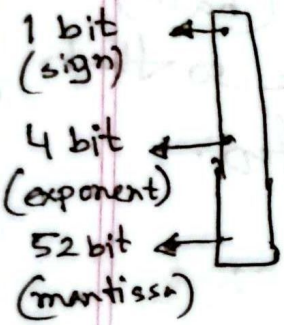
So, $\delta_{max} = \varepsilon_m = \frac{\frac{1}{2}\beta^{-m} \times \beta^e}{\beta^{-1}\beta^e} = \frac{1}{2}\beta^{1-m}$

Class-3

IEEE format

Double Precision format → 64 bit.

So, Among 64 bits,



sign (+/-) → 1 bit

exponent → 4 bit

mantissa → 52 bit

Starting with a normalized form.

$$\pm (1.d_1 d_2 d_3 \cdots d_m)_\beta \times \beta^e$$

So, we can write,

$$1 \leftarrow \pm \underbrace{(1.d_1 d_2 d_3 \cdots d_{52})}_{52} \beta \times \beta^{\overbrace{e}^{11}}$$

Here 11 bit means 2^{11} combination $= 2048$

$$\text{So, } e \in [0, (2^{11}-1)]$$

$$\text{So, } e \in [0, 2047]$$

$$\text{So, max} = +(1.1111\cdots1)_2 \times 2^{2047} = \text{Very large value}$$

$$\text{min} = +(1.0000\cdots1)_2 \times 2^0 = 1$$

Exponential Biased

আগে we got a very large value But lowest value was 0. so we'll shift some value from the positive side (half) that'll be shifted to the left side so that we can get the lowest value too.



Here, total number = 2048

$$\text{middle value} = \frac{2048}{2} = 1024$$

Removing an index (as 0 counts too) $= (1024 - 1) = 1023$

So, $\pm (1.d_1 d_2 d_3 \ldots d_{52})_\beta \times \beta^{e-1023}$

So, $e - 1023 \in [-1023 - 1024]$

So,

$max = +(1.111\ldots1)_2 \times 2^{1024} \rightarrow \pm\infty$

$min = +(1.000\ldots0)_2 \times 2^{-1023} \rightarrow \pm 0$

- Thus we can represent both large number and small numbers.